TOYOTA MANUFACTURING PROCESS IMPROVEMENT

Using Lean Six Sigma and Machine Learning

Project Report — Draft | Mohamed Riyaz S. | VIT Chennai

Abstract

This educational project combines Lean Six Sigma DMAIC, statistical process analysis and machine learning to explore manufacturing variation and defect risk. It establishes a baseline, investigates patterns through Pareto and stratified analysis, trains a Random Forest defect-risk classifier, and organizes improvement and control activities in a dashboard. The bundled dataset is synthetic and is not Toyota production data. Company-specific claims require authorized real data and independent validation.

1. Introduction

Manufacturing quality improvement requires reliable measurements, a clear defect definition and disciplined investigation of process variation. DMAIC provides a structured method to define customer-critical requirements, measure current performance, analyze potential causes, test improvements and sustain verified gains. Machine learning can complement this process by identifying predictive patterns, but cannot establish causation on its own.

2. Problem Statement

Quality teams may need to combine process measurements across machines, shifts, lines and material batches. This project develops a reproducible workflow to establish a baseline, prioritize defect patterns, explore process variation, estimate defect risk and organize improvement/control activities. It does not claim access to Toyota internal records or that a Toyota production process was changed.

3. Objectives

- Apply DMAIC to a manufacturing quality case study.
- Establish baseline defect and process KPIs.
- Use Pareto and statistical analysis to explore patterns.
- Train and evaluate a Random Forest defect-risk classifier.
- Propose evidence-led improvement and control-plan elements.
- Present results through a Streamlit dashboard.

4. Define and Measure

Define the problem, scope, CTQs, goal, stakeholders, SIPOC and project charter. Measure using an operational defect definition, a data collection plan, valid units, sampling, measurement-system adequacy, missingness checks and baseline performance. The demonstration schema includes line, shift, machine, operator experience, temperature, pressure, cycle time, vibration, tool wear, material batch and defect label.

5. Analyze

Use Pareto to prioritize defect categories or machines; stratify by line, shift, machine and material; inspect distributions and correlations; and conduct statistical tests only when assumptions and sampling design are appropriate. Feature importance is predictive association, not proof of root cause. Engineering validation is required before changing process settings.

6. Improve and Control

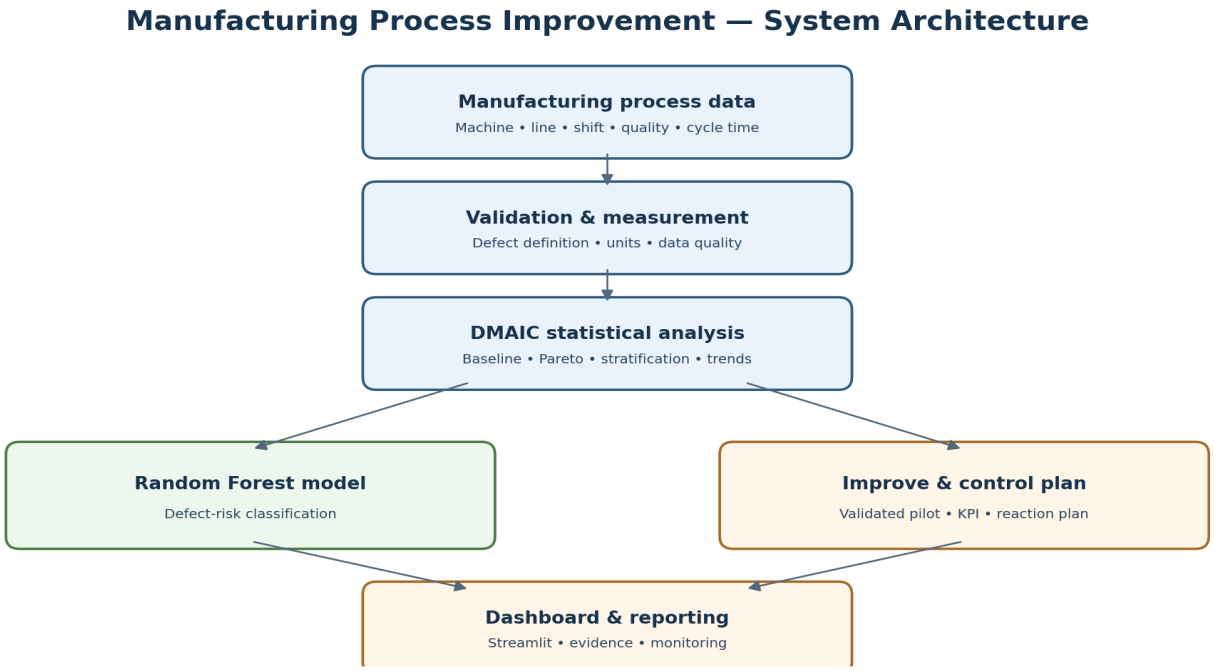
Select countermeasures from verified root causes, pilot under an approved protocol, and evaluate before/after outcomes using an appropriate design. The Control phase defines KPI owners, sampling frequency, control charts, standard work, audits and reaction plans. Control limits are estimated from process behavior; specification limits are engineering/customer requirements.

7. Machine Learning

A Random Forest classifier estimates defect probability from numeric process measurements and categorical line, shift and machine fields. Preprocessing is fitted inside a scikit-learn Pipeline. Metrics include precision, recall, F1, ROC-AUC and confusion matrix. Threshold selection should reflect the cost of false negatives and false positives. Prediction is not causal evidence.

8. System Architecture

Proposed workflow; not a live Toyota production integration.



9. Results and Discussion

Insert actual baseline KPIs, Pareto findings, model metrics, validation design and control-chart interpretation after running on authorized data. Synthetic outputs only test code and dashboard behavior. Do not report synthetic values as Toyota performance, defect reduction, sigma improvement, savings or verified root causes.

10. Limitations and Future Work

Limitations include synthetic data, illustrative variables and lack of factory measurement-system study. Future work can incorporate authorized production data, Gage R&R;, temporal validation, cost-sensitive thresholds, explainability, designed experiments, pilot studies and a sustained control plan.

11. Conclusion and References

The project demonstrates a reproducible DMAIC, statistical and machine-learning workflow. Practical value depends on valid measurements, an agreed defect definition, appropriate assumptions, engineering review and verified trials. Add exact citations for data, Lean Six Sigma references, statistical methods, libraries and any Toyota public materials actually used. Do not imply Toyota endorsement.